

SmartRecSys End-to-End Hybrid Recommendation & Explainability Pipeline

1. STUDENT ACADEMIC PROFILE INTAKE

Department | Degree Program (B.Tech 4-Yr / BCA 3-Yr) | Batch Year & Active Semester | Target Domain Specialties
Prior Completed Prerequisites | Certified Skill Competencies | Declared Career Aspiration

Resolves Zero-History Freshman Cold-Start via Multi-Token Semantic Formulation

Profile Tokens

User/Course ID

Historical Links

2A. SEMANTIC PROFILING

Sublinear TF-IDF Vectorizer
Vocab Dimension: $|V|$ tokens
Continuous Cosine Metric
 $S_{sem}(u, i)$ in $[0, 1]$

Content Continuity Bridge

2B. DEEP LEARNING (NCF)

PyTorch Dual Embeddings ($d=16$)
Concatenation: $z_0 = [P_u || Q_i]$
3-Layer Non-Linear MLP (ReLU, Dropout)
Sigmoid Prediction: $S_{neu}(u, i)$

Accelerated via Apple Silicon MPS / CUDA

2C. SVD FACTORIZATION

Truncated SVD Matrix Decomp.
Rank $k=12$ Latent Curricular Factors
Frobenius Norm Reconstruction
 $R_{hat} = U * \Sigma * V^T$

Historical Campus Interaction Matrix

3. PEDAGOGICAL POLICY & DOMAIN AFFINITY MASKING

Domain Affinity Mask S_{dom} : Strict Boundary Guard ($S_{dom} = 0 \rightarrow S_{final} = 0.05$ Penalty)
Cognitive Difficulty Alignment S_{dif} : $\Delta(Target_Level, Course_Level)$ in $[0.10, 1.00]$
Prerequisite Verification: Chaining completed foundations to prospective syllabi
Eliminates Out-of-Domain Leakage & Prevents Cognitive Overload

Regularized Weights

Attribution Features

4. MASTER CALIBRATED HYBRID FUSION

$S_{raw} = 0.40*S_{dom} + 0.30*S_{sem} + 0.20*S_{dif} + 0.10*S_{neu}$
 $Score_{final} = \min(0.98, \max(0.65, 0.65 + 0.33*S_{raw}))$
Top-K Academic Ranking Permutation
+16.6% Recall@5 Gain over CF

Ranked Permutation

5. XAI EXPLAINABILITY & AUDIT REPORT

Pedagogical Attribution: $\psi_{dom} + \psi_{career} + \psi_{prereq}$
Persistent SQLite Logging (smartrecsys.db)
Interactive UI Cards & One-Click Printable PDF Advisory
Institutional Transparency & Advisory Compliance